



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO KANAGAMANI AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

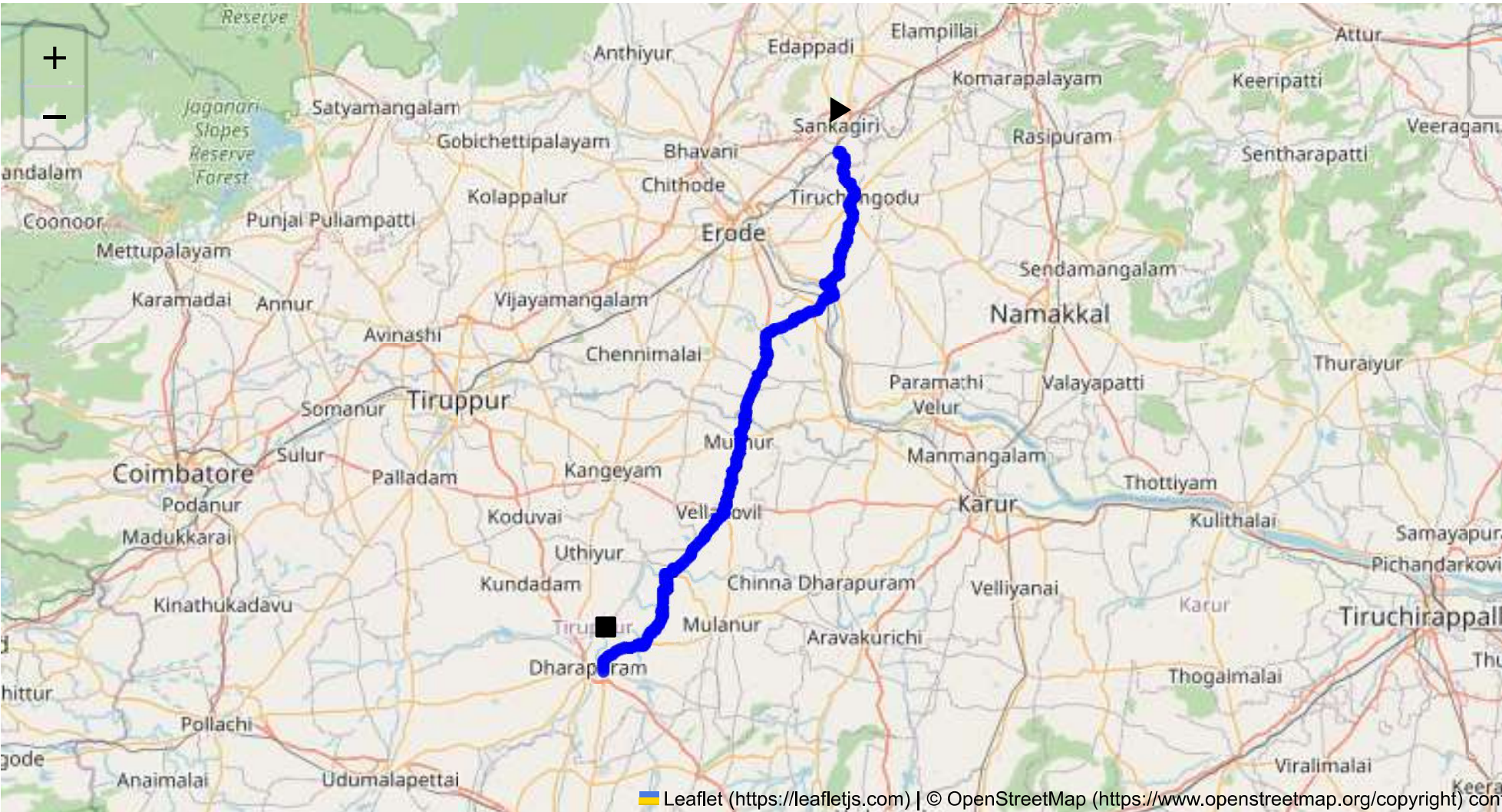
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 101.18 km
Estimated Duration: 2.4 hours
Adjusted Duration (Heavy Vehicle): 3.0 hours
Start: (11.4381, 77.8734)
End: (10.731541, 77.546519)



Welcome to the Journey Risk Management Study

Route Safety Analysis Report

1. Overview of the Route Map

The route from CVQF+23W, Sangagiri to PGJW+PGQ, Upputhurai Palayam traverses approximately 101.18 kilometers through the Tamil Nadu state. The main connecting roads include National and State

Highways, which are frequently used by both passenger vehicles and heavy trucks. The journey covers both rural and semi-urban areas.

2. Typical Weather Conditions and Potential Weather-Related Hazards

The region typically experiences tropical climate conditions. The summer months can be extremely hot, affecting vehicle performance. Monsoon season, primarily from June to September, may bring heavy rains, potentially leading to waterlogging and reduced visibility. Fog can be a concern during winter mornings.

3. Analysis of Traffic Patterns

Traffic congestion is primarily observed during local commuter peak hours: 8 AM to 10 AM and 5 PM to 8 PM. Congestion tends to increase near towns and market areas. Limited bypass options around busy junctions may exacerbate traffic delays in the area.

4. Assessment of Road Quality and Infrastructure

The quality of roads varies across the route. Highways are generally well-maintained, but secondary roads may have potholes and uneven surfaces, especially post-monsoon. Signage and road markings may occasionally be unclear, necessitating cautious driving.

5. Suggestions for Alternative Routes for Emergencies

If an emergency detour is necessary, National Highway 44 provides an alternate north-south axis. However, any diversion should be verified for local traffic conditions and suitability for heavy vehicles, as not all secondary roads are built to accommodate large trucks.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Transporting hazardous materials requires adherence to state and national regulations, including necessary permits and display of appropriate hazard placards. Specific routes may be restricted for such transport at certain times or in high-density areas.

7. Overview of Historical Incidents

While direct data on incidents involving hazardous materials is not extensive, reports indicate road mishaps are mostly due to poor visibility, overloaded cargo, and driver fatigue. Regular checks on vehicle condition and adherence to rest protocols are advised.

8. Environmental Considerations and Sensitive Areas

The route passes through agricultural zones, and care should be taken to prevent contamination. Spillage of materials could have detrimental effects on local water bodies and soil, impacting agriculture and local communities.

9. Analysis of Communication Coverage

The route generally maintains reliable communication network coverage, with possible dead zones in more rural sections. It is advisable to be equipped with additional communication devices like a two-way radio.

10. Estimated Emergency Response Times

Response times can vary but typically range between 30 to 60 minutes in more populated areas, extending to over an hour in rural regions. Establishing contact with local emergency services and planning for self-rescue protocols may be prudent.

12. Overall Summary of Risk Assessment

Overall, the route is moderately risky due to a mix of road quality, weather impacts, and traffic patterns. Preparedness for adverse weather, maintaining up-to-date vehicle servicing, and awareness of local regulations are key to mitigating risks. Ensuring proper rest and adherence to driving hours can significantly enhance safety on this route.

In conclusion, while the route has challenges typical of regional highways, observance of safety protocols can substantially mitigate these risks, ensuring a safer journey for trucks carrying hazardous materials.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Roundabout	High	11.04637, 77.73761	15 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44022, 77.87546	15 KM/Hr
4	Turn	Medium	11.43107, 77.88171	30 KM/Hr
5	Turn	Medium	11.39976, 77.88382	30 KM/Hr
6	Turn	Medium	11.39690, 77.88852	30 KM/Hr
7	Turn	High	11.33268, 77.88827	15 KM/Hr
8	Turn	Medium	11.33165, 77.88697	30 KM/Hr
9	Turn	High	11.32130, 77.88484	15 KM/Hr
10	Turn	Medium	11.32102, 77.88435	30 KM/Hr
11	Turn	Medium	11.31183, 77.88094	30 KM/Hr
12	Turn	Medium	11.31151, 77.88008	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
13	Turn	Medium	11.30805, 77.87934	30 KM/Hr
14	Turn	Medium	11.30580, 77.87761	30 KM/Hr
15	Turn	Medium	11.29748, 77.87673	30 KM/Hr
16	Turn	Medium	11.29668, 77.87515	30 KM/Hr
17	Turn	Medium	11.29549, 77.87464	30 KM/Hr
18	Turn	Medium	11.29417, 77.87309	30 KM/Hr
19	Turn	Medium	11.27096, 77.87034	30 KM/Hr
20	Turn	Medium	11.27057, 77.86827	30 KM/Hr
21	Turn	High	11.26028, 77.85633	15 KM/Hr
22	Turn	High	11.25791, 77.85522	15 KM/Hr
23	Turn	Medium	11.25287, 77.86220	30 KM/Hr
24	Turn	Medium	11.24410, 77.86671	30 KM/Hr
25	Turn	High	11.24365, 77.86668	15 KM/Hr
26	Turn	High	11.24100, 77.86030	15 KM/Hr
27	Turn	High	11.24193, 77.85850	15 KM/Hr
28	Turn	Medium	11.22478, 77.84241	30 KM/Hr
29	Turn	Medium	11.21398, 77.81408	30 KM/Hr
30	Turn	Medium	11.20949, 77.81173	30 KM/Hr
31	Turn	Medium	11.18875, 77.77206	30 KM/Hr
32	Blind Spot	Blind Spot	11.18861, 77.77108	10 KM/Hr
33	Turn	Medium	11.16466, 77.77354	30 KM/Hr
34	Turn	Medium	11.16321, 77.77438	30 KM/Hr
35	Turn	High	11.13693, 77.76539	15 KM/Hr
36	Turn	High	11.04545, 77.73638	15 KM/Hr
37	Turn	High	10.99038, 77.72518	15 KM/Hr
38	Turn	High	10.98970, 77.72385	15 KM/Hr
39	Turn	Medium	10.96614, 77.71685	30 KM/Hr
40	Turn	High	10.94597, 77.71319	15 KM/Hr
41	Turn	High	10.94582, 77.71257	15 KM/Hr
42	Turn	Medium	10.94256, 77.71174	30 KM/Hr
43	Turn	Medium	10.93834, 77.70347	30 KM/Hr
44	Turn	Medium	10.92356, 77.69226	30 KM/Hr
45	Turn	High	10.87230, 77.64953	15 KM/Hr
46	Turn	High	10.87207, 77.64474	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
47	Turn	Medium	10.87110, 77.64415	30 KM/Hr
48	Turn	High	10.86447, 77.64345	15 KM/Hr
49	Turn	Medium	10.86191, 77.63340	30 KM/Hr
50	Turn	High	10.85748, 77.63339	15 KM/Hr
51	Turn	High	10.85691, 77.63242	15 KM/Hr
52	Turn	Medium	10.85306, 77.63348	30 KM/Hr
53	Turn	Medium	10.85273, 77.63395	30 KM/Hr
54	Turn	Medium	10.84087, 77.63395	30 KM/Hr
55	Turn	High	10.83995, 77.62962	15 KM/Hr
56	Turn	High	10.83832, 77.62940	15 KM/Hr
57	Turn	Medium	10.83770, 77.62869	30 KM/Hr
58	Turn	Medium	10.81734, 77.62883	30 KM/Hr
59	Turn	Medium	10.80957, 77.62858	30 KM/Hr
60	Turn	Medium	10.79001, 77.61948	30 KM/Hr
61	Turn	Medium	10.78284, 77.61251	30 KM/Hr
62	Turn	High	10.76663, 77.60179	15 KM/Hr
63	Turn	Medium	10.76745, 77.59641	30 KM/Hr
64	Turn	High	10.74735, 77.54864	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
6	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
7	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
9	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
10	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium
14	hospital	Government Hospital, Elumathur	11.18533, 77.772496	30 km/h	Medium
15	hospital	Malar Hospital, Muthur	11.0438049, 77.7361156	30 km/h	Medium
16	hospital	SIBI Hospital, Vellakoil	10.9569141, 77.7160216	30 km/h	Medium
17	hospital	KG Hospital, Vellakoil	10.9511, 77.714116	30 km/h	Medium
18	hospital	Vellakoil Government Hospital	10.9410661, 77.7110372	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.04637, 77.73761



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44022, 77.87546



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43107, 77.88171



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.39976, 77.88382



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.39690, 77.88852



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.33268, 77.88827



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.33165, 77.88697



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.32130, 77.88484



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.32102, 77.88435



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.31183, 77.88094



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.31151, 77.88008



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.30805, 77.87934



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.30580, 77.87761



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.27096, 77.87034



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.27057, 77.86827



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.26028, 77.85633



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.25791, 77.85522



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.25287, 77.86220



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.24410, 77.86671



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.24365, 77.86668



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.24100, 77.86030



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.24193, 77.85850



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.22478, 77.84241



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.21398, 77.81408



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.20949, 77.81173



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.18875, 77.77206



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.18861, 77.77108



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.16466, 77.77354



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.16321, 77.77438



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.13693, 77.76539



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.04545, 77.73638



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.99038, 77.72518



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.98970, 77.72385



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.96614, 77.71685



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.94597, 77.71319



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.94582, 77.71257



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.94256, 77.71174



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.93834, 77.70347



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.92356, 77.69226



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.87230, 77.64953



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.87207, 77.64474



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.87110, 77.64415



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.86447, 77.64345



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.86191, 77.63340



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.85748, 77.63339



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.85691, 77.63242



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.85306, 77.63348



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.85273, 77.63395



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.84087, 77.63395



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.83995, 77.62962



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.83832, 77.62940



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.83770, 77.62869



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.81734, 77.62883



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.80957, 77.62858



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.79001, 77.61948



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.78284, 77.61251



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.76663, 77.60179



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.76745, 77.59641



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.74735, 77.54864

